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CASE STUDY REPORT

TOPIC: URBAN AIR QUALITY IN INDIAN CITIES – EXPLORATORY DATA ANALYSIS AND HYPOTHESIS TESTING

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PROFESSIONAL ELECTIVE -I
DATA SCIENCE

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URBAN AIR QUALITY IN INDIAN CITIES – EXPLORATORY DATA ANALYSIS AND HYPOTHESIS TESTING

ABSTRACT

Air pollution is one of the major environmental challenges affecting urban areas in India. Rapid urbanisation, industrialisation, increasing transportation activities and population growth contribute significantly to the deterioration of air quality. Monitoring and analysing air pollution data is essential for understanding environmental conditions and supporting sustainable urban development.

This project, titled **"Urban Air Quality in Indian Cities – Exploratory Data Analysis and Hypothesis Testing,"** focuses on analysing historical air quality data collected from Indian cities between 2015 and 2020. The study primarily considers two important air pollutants: Particulate Matter 2.5 (PM2.5) and Nitrogen Dioxide (NO_2).

The project uses the publicly available Central Pollution Control Board (CPCB) air quality dataset. Python-based data analytics libraries, including Pandas, NumPy, SciPy, Statsmodels, Matplotlib and Seaborn, are used for data preprocessing, exploratory data analysis and statistical evaluation.

The analysis examines pollutant distributions across cities and seasons, calculates descriptive statistics and confidence intervals, and investigates differences between the corresponding periods of 2019 and 2020 during the COVID-19 lockdown. Welch's independent t-test and the Mann–Whitney U test are applied to evaluate statistical differences. Pearson and Spearman correlation techniques are used to investigate relationships between air pollutants.

The project demonstrates how statistical analysis and data visualisation can support environmental research, identify pollution patterns and contribute to sustainable urban planning. It also highlights the importance of distinguishing statistical correlation from causal relationships.

Keywords: Air Pollution, PM2.5, NO_2, Exploratory Data Analysis, Hypothesis Testing, Confidence Interval, Correlation Analysis, CPCB, Python, Environmental Data Analytics.

CHAPTER 1: INTRODUCTION

1.1 Background

Air pollution is a serious environmental and public health concern in many Indian cities. Increasing population, industrial development, construction activities and transportation contribute to the release of harmful pollutants into the atmosphere.

Air pollutants affect human health, environmental quality and climate conditions. Long-term exposure to particulate matter and gaseous pollutants can increase the risk of respiratory and cardiovascular diseases.

With the development of environmental monitoring systems, large amounts of air quality data are collected from monitoring stations across India. Analysing these datasets using statistical and computational techniques provides valuable information about pollution patterns.

Exploratory Data Analysis (EDA) helps researchers understand the characteristics of datasets, identify unusual observations and discover relationships between variables.

Hypothesis testing provides a statistical framework for determining whether observed differences between datasets are significant.

This project applies these techniques to historical Indian urban air quality data.

1.2 Problem Statement

Air pollution concentrations vary across cities, seasons and time periods. However, identifying these variations requires systematic analysis of historical environmental data.

The COVID-19 lockdown introduced significant restrictions on transportation, industrial activities and public movement in 2020. These restrictions may have influenced urban air pollution concentrations.

The primary problem addressed in this project is to analyse PM2.5 and NO_2 concentrations across selected Indian cities and determine whether pollutant levels during the 2020 lockdown period differed significantly from the corresponding period in 2019.

The project also examines seasonal pollution patterns, estimates confidence intervals and investigates correlations between pollutants.

1.3 Objectives

The main objectives are:

1. To collect historical CPCB air quality data for Indian cities from 2015 to 2020.
2. To perform data cleaning and preprocessing.
3. To analyse PM2.5 and NO_2 distributions.
4. To compare pollution levels across different cities.
5. To investigate seasonal variations in air pollution.
6. To calculate confidence intervals for pollutant concentrations.
7. To compare corresponding lockdown-period observations from 2019 and 2020.
8. To perform Welch's t-test and Mann–Whitney U test.
9. To identify relationships between different pollutants using correlation analysis.
10. To interpret statistical findings in the context of environmental sustainability.

1.4 Scope of the Project

The project focuses on historical air quality observations collected from selected Indian cities.

The study covers:

- Time period: 2015–2020.
- Primary pollutants: PM2.5 and NO_2.
- Additional pollutants: PM10, SO_2, CO and O_3, where available.
- Spatial analysis: City-wise comparison.
- Temporal analysis: Year-wise and seasonal comparison.
- Statistical analysis: Confidence intervals and hypothesis testing.
- Correlation analysis: Pearson and Spearman correlation.

The study is limited to the available observations in the selected dataset.

1.5 Significance of the Project

This project contributes to environmental data analytics by demonstrating how statistical computing can be used to understand urban air quality.

The findings can help environmental researchers, policymakers and urban planners understand pollution patterns and identify areas requiring further investigation.

The project supports:

- SDG 11 – Sustainable Cities and Communities.
- SDG 13 – Climate Action.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Literature review is an important research activity that examines previous studies, scientific methods and existing developments related to the selected research topic.

Air quality monitoring and statistical analysis have become important research areas due to increasing environmental concerns.

2.2 Air Quality Monitoring

Air quality monitoring involves collecting measurements of pollutants from monitoring stations.

The Central Pollution Control Board monitors air quality across different locations in India.

These measurements provide information about pollutant concentrations and help identify environmental trends.

2.3 Particulate Matter Analysis

PM2.5 represents fine particulate matter with an aerodynamic diameter of 2.5 micrometres or less.

Research on particulate matter focuses on its concentration, distribution, health impacts and seasonal variations.

PM2.5 concentrations can be influenced by transportation, industrial activities, combustion, atmospheric conditions and regional pollution transport.

2.4 Nitrogen Dioxide Analysis

Nitrogen dioxide is an important atmospheric pollutant associated with fuel combustion.

Major sources include:

- Road transportation.
- Industrial combustion.
- Power generation.
- Other fossil-fuel combustion activities.

NO_2 monitoring helps researchers understand pollution associated with urban activities.

2.5 Exploratory Data Analysis

EDA is widely used in environmental research to examine data characteristics.

Common EDA techniques include:

- Descriptive statistics.
- Histograms.
- Boxplots.
- Scatter plots.
- Correlation heatmaps.
- Time-series graphs.

These techniques help identify patterns and potential anomalies before formal statistical analysis.

2.6 Statistical Hypothesis Testing

Hypothesis testing is used to evaluate whether observed differences between samples provide sufficient statistical evidence against a null hypothesis.

The independent t-test is commonly used for comparing two sample means.

The Mann–Whitney U test provides a rank-based alternative for comparing two independent samples.

Both techniques can contribute to environmental data analysis when their assumptions and limitations are considered.

2.7 Research Gap

Although historical air quality measurements are available, analysing the data through multiple statistical perspectives is important.

This project combines city-wise analysis, seasonal analysis, confidence interval estimation, lockdown-period comparison and pollutant correlation analysis in a single analytical framework.

It also emphasises the limitations of observational comparisons and the distinction between correlation and causation.

CHAPTER 3: DATASET DESCRIPTION AND TECHNOLOGIES

3.1 Dataset Description

The project uses publicly available CPCB city-day air quality observations covering 2015–2020.

The dataset provides pollutant measurements collected through air quality monitoring systems.

The exact dataset structure depends on the downloaded source files.

3.2 Dataset Attributes

Attribute	Description
City	Name of the city
Date	Date of measurement
PM2.5	Fine particulate concentration
PM10	Particulate matter concentration
NO_2	Nitrogen dioxide concentration
SO_2	Sulphur dioxide concentration
CO	Carbon monoxide concentration
O_3	Ozone concentration

PM2.5, PM10, NO_2, SO_2 and O_3 are commonly expressed in $\mu g/m^3$. CO may use a different unit depending on the source.

3.3 Data Source

Primary source:

Central Pollution Control Board, Government of India.

Website: https://cpcb.nic.in/

Additional source:

Open Government Data Platform India.

Website: https://www.data.gov.in/

3.4 Technologies Used

Python

Python is the primary programming language used for the implementation.

Pandas

Pandas is used for loading, cleaning, filtering, grouping and analysing the dataset.

NumPy

NumPy supports numerical calculations and mathematical operations.

SciPy

SciPy provides statistical functions for hypothesis testing.

Statsmodels

Statsmodels is used for confidence interval calculations and statistical modelling.

Matplotlib

Matplotlib is used to generate graphs and visual representations.

Seaborn

Seaborn provides statistical visualisation techniques such as heatmaps, histograms and boxplots.

Jupyter Notebook

Jupyter Notebook provides an interactive environment for implementing and documenting the analysis.

CHAPTER 4: PROPOSED METHODOLOGY

4.1 Overview

The proposed methodology consists of data collection, preprocessing, exploratory data analysis, confidence interval estimation, hypothesis testing and correlation analysis.

4.2 System Workflow

The proposed workflow is:

1. Collect CPCB air quality data.
2. Load the dataset into Python.
3. Perform data preprocessing.
4. Calculate descriptive statistics.
5. Analyse city-wise pollution distributions.
6. Analyse seasonal variations.
7. Estimate confidence intervals.
8. Compare 2019 and 2020 lockdown-period observations.
9. Perform statistical hypothesis testing.
10. Analyse pollutant correlations.
11. Generate visualisations.
12. Interpret and document the results.

4.3 Data Preprocessing

Data preprocessing is necessary to improve the quality and reliability of statistical analysis.

The following activities are performed:

Missing-value identification: Identify missing pollutant measurements.

Duplicate removal: Remove duplicate observations based on appropriate record identifiers.

Date conversion: Convert date values into Python datetime format.

Data type conversion: Convert pollutant measurements into numerical values.

Invalid-value handling: Identify negative values and other invalid measurements.

City standardisation: Ensure consistent city names.

Missing observations should be handled carefully to avoid introducing bias.

4.4 Exploratory Data Analysis

EDA is performed to understand the dataset.

4.4.1 Descriptive Statistics

The following statistical measures are calculated:

- Mean.
- Median.
- Standard deviation.
- Minimum.
- Maximum.
- First quartile.
- Third quartile.

These measures help understand pollutant distributions and variability.

4.4.2 City-wise Analysis

The mean PM2.5 and NO_2 concentrations are calculated separately for each city.

The results are visualised using bar charts.

This analysis helps identify differences in pollutant concentrations across selected cities.

4.4.3 Seasonal Analysis

The observations are divided into four seasons:

Season	**Months**
Winter	December–February
Summer	March–May
Monsoon	June–September
Post-monsoon	October–November

Seasonal average pollutant concentrations are calculated and compared using graphs.

4.4.4 Distribution Analysis

Histograms and boxplots are generated to understand the distribution of pollutant measurements.

These visualisations help identify skewness, variability and potential outliers.

4.5 Confidence Interval Estimation

A confidence interval estimates a range of plausible values for a population parameter.

The 95% confidence interval for the sample mean is calculated using:

$CI = \bar{x} \pm t \times (s / \sqrt{n})$

Where:

- $\bar{x}$ represents the sample mean.
- t represents the critical t-value.
- s represents the sample standard deviation.
- n represents the sample size.

Confidence intervals are calculated for PM2.5 and NO_2.

The results are interpreted to understand the uncertainty associated with average pollutant concentrations.

4.6 Hypothesis Testing

The hypothesis-testing component investigates whether air pollution concentrations differed between the corresponding periods of 2019 and 2020.

The primary lockdown comparison period is 25 March to 31 May.

Null Hypothesis

H_0: There is no significant difference in pollutant concentrations between the 2019 comparison period and the 2020 lockdown period.

Alternative Hypothesis

H_1: There is a significant difference in pollutant concentrations between the two periods.

The significance level is set to 0.05.

4.6.1 Welch's Independent T-Test

Welch's t-test compares the means of two independent groups without assuming equal population variances.

The test is applied separately to PM2.5 and NO_2.

If the p-value is below 0.05, the null hypothesis is rejected.

4.6.2 Mann–Whitney U Test

The Mann–Whitney U test is a non-parametric statistical test that compares two independent sample distributions using ranks.

It is useful when data distributions are skewed or contain extreme observations.

The results are compared with the t-test findings to understand whether the conclusions are consistent.

4.6.3 Statistical Considerations

Daily observations may be temporally correlated. Therefore, the analysis should consider city-level aggregation, matching dates and dependence between observations.

Where appropriate, city-specific comparisons and paired statistical methods should be considered.

The results should not automatically be interpreted as evidence that lockdown restrictions caused the observed changes.

4.7 Correlation Analysis

Correlation analysis examines the relationship between pollutant measurements.

Pearson Correlation

Pearson correlation measures the strength and direction of a linear relationship between two numerical variables.

Spearman Correlation

Spearman correlation measures monotonic relationships using ranked observations.

The correlation matrix is visualised using a heatmap.

4.8 Correlation Does Not Imply Causation

A positive correlation between PM2.5 and NO_2 does not prove that one pollutant causes an increase in the other.

Both pollutants may be influenced by common factors such as:

- Vehicle emissions.
- Industrial activities.
- Weather conditions.
- Atmospheric mixing.
- Seasonal changes.

Therefore, correlation analysis identifies statistical associations rather than direct causal relationships.

CHAPTER 5: EXPERIMENTAL IMPLEMENTATION

5.1 Development Environment

The project is implemented using Python and Jupyter Notebook.

Required libraries:

- pandas
- numpy
- scipy
- statsmodels
- matplotlib
- seaborn

5.2 Dataset Loading

The dataset is loaded using the Pandas library.

The dataset structure, column names and data types are examined.

5.3 Data Cleaning

Missing observations, duplicates, invalid measurements and date formats are checked and handled.

5.4 Statistical Analysis

Descriptive statistics, confidence intervals and hypothesis tests are implemented using Python libraries.

5.5 Visualisation

The following graphs are generated:

1. PM2.5 distribution histogram.
2. NO_2 distribution histogram.
3. City-wise pollution bar chart.
4. Seasonal pollution boxplot.
5. Correlation heatmap.
6. 2019 versus 2020 pollutant comparison.
7. Year-wise pollution trend graph.

5.6 Experimental Results

The actual results should be populated after executing the implementation on the CPCB dataset.

City-wise Analysis

City	Mean PM2.5	Mean NO_2
City 1	To be calculated	To be calculated
City 2	To be calculated	To be calculated
City 3	To be calculated	To be calculated

Seasonal Analysis

Season	Mean PM2.5	Mean NO_2
Winter	To be calculated	To be calculated
Summer	To be calculated	To be calculated
Monsoon	To be calculated	To be calculated
Post-monsoon	To be calculated	To be calculated

Confidence Interval Results

Pollutant	Mean	Lower CI	Upper CI
PM2.5	To be calculated	To be calculated	To be calculated
NO_2	To be calculated	To be calculated	To be calculated

Hypothesis Testing Results

Pollutant	Statistical Test	Test Statistic	P-value
PM2.5	Welch t-test	To be calculated	To be calculated
PM2.5	Mann–Whitney U	To be calculated	To be calculated
NO_2	Welch t-test	To be calculated	To be calculated
NO_2	Mann–Whitney U	To be calculated	To be calculated

Correlation Results

The correlation matrix should be generated using the available pollutant columns.

The strongest positive and negative relationships should be discussed based on the actual calculated values.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Results

The project evaluates the distribution of air pollutants across different cities and seasons.

Descriptive statistics provide an understanding of average concentrations and variations.

Confidence intervals provide uncertainty estimates for pollutant means.

Hypothesis testing evaluates whether the corresponding 2019 and 2020 observations show statistically significant differences.

Correlation analysis identifies associations between pollutant measurements.

6.2 Discussion

The city-wise analysis is used to understand geographical differences in pollution levels.

The seasonal analysis examines how pollutant concentrations change during winter, summer, monsoon and post-monsoon periods.

The lockdown comparison evaluates the statistical evidence of differences between two corresponding periods.

The correlation analysis helps identify potential common pollution patterns.

The results should be interpreted considering monitoring coverage, missing data, temporal dependence and meteorological influences.

6.3 Key Findings

The final findings should be completed after analysing the dataset.

The following research questions should be answered:

1. Which selected city has the highest average PM2.5 concentration?
2. Which season has the highest PM2.5 concentration?
3. Are NO_2 concentrations different across cities?
4. Is there a statistically significant difference between the 2019 and 2020 comparison periods?
5. Which pollutants show the strongest correlation?
6. What are the limitations of interpreting correlation as causation?

CHAPTER 7: LIMITATIONS AND FUTURE ENHANCEMENTS

7.1 Limitations

1. Missing pollutant observations may affect the analysis.
2. Monitoring station coverage may vary between cities.
3. Weather conditions may influence pollutant concentrations.
4. Daily observations may not be statistically independent.
5. The 2019–2020 comparison cannot independently establish the causal effect of lockdown.
6. Some pollutants may not be available consistently across all locations.

7.2 Future Enhancements

The project can be extended through:

1. Development of an interactive dashboard using Streamlit.
2. Integration of temperature, humidity, rainfall and wind-speed data.
3. Implementation of regression models.
4. Air quality forecasting using time-series methods.
5. Machine learning-based pollution classification.
6. Extension of the study to additional Indian cities.
7. Application of Difference-in-Differences analysis to investigate lockdown-associated changes.

CHAPTER 8: CONCLUSION

This project presents an analytical framework for understanding urban air quality in Indian cities using historical CPCB data from 2015 to 2020.

The study uses exploratory data analysis, descriptive statistics, confidence interval estimation, hypothesis testing and correlation analysis.

City-wise and seasonal analysis provides insights into spatial and temporal variations in PM2.5 and NO_2 concentrations.

Welch's t-test and Mann–Whitney U test are applied to investigate statistical differences between the corresponding 2019 and 2020 lockdown-period observations.

Correlation analysis identifies relationships between pollutants and demonstrates the importance of distinguishing association from causation.

The proposed methodology provides a foundation for environmental data analytics and can be extended using machine learning, forecasting and interactive visualisation techniques.

Overall, the project demonstrates how statistical computing can support environmental monitoring and sustainable urban development, contributing to Sustainable Development Goals 11 and 13.

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